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MODULAR INVERTER AIR-COOLED CHILLER (HEAT PUMP)

CONTENTS

PART 1 Features of Modular Inverter Units

PART 2 Analysis of Competitive Products

PART 1

Features of Modular Inverter Units

Patented design

Patent No.: 202030284100.8



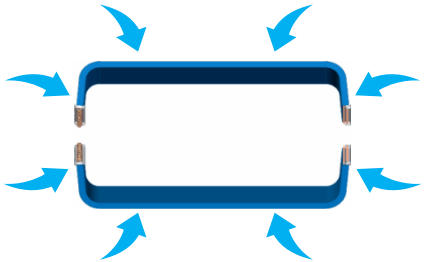
01 Upgraded appearance

Fan, compressor, and electric control box are concealed to ensure safety and consistent aesthetic effect

Four-way air return

02

Dual U-shaped heat exchanger, windward area increased by 45%, smooth air return from all sides



03 Uniformly coated vertical frame

Sheet metal parts are electrostatically sprayed to give them an elegant look and corrosion resistance feature

Advanced 25HP DC inverter scroll compressor

Mitsubishi patent

Scroll track sealing ring technology

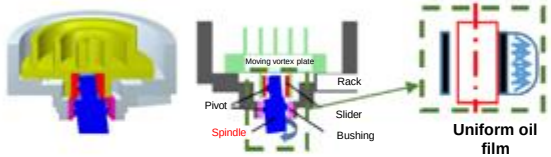
Reduce leakage during scroll compression to increase overall system efficiency while avoiding rigid collisions of moving parts to improve overall system life.



Mitsubishi patent

Pivot bearing design

Under heavy loads, more uniform bearing gap, smoother compressor operation and lower noises for larger operating range, ensuring the life of the overall system.

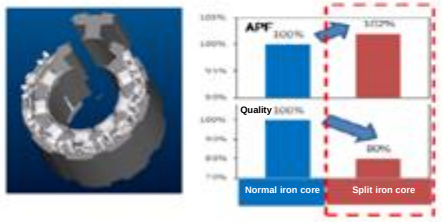


(Patent: 2930046)

Mitsubishi patent

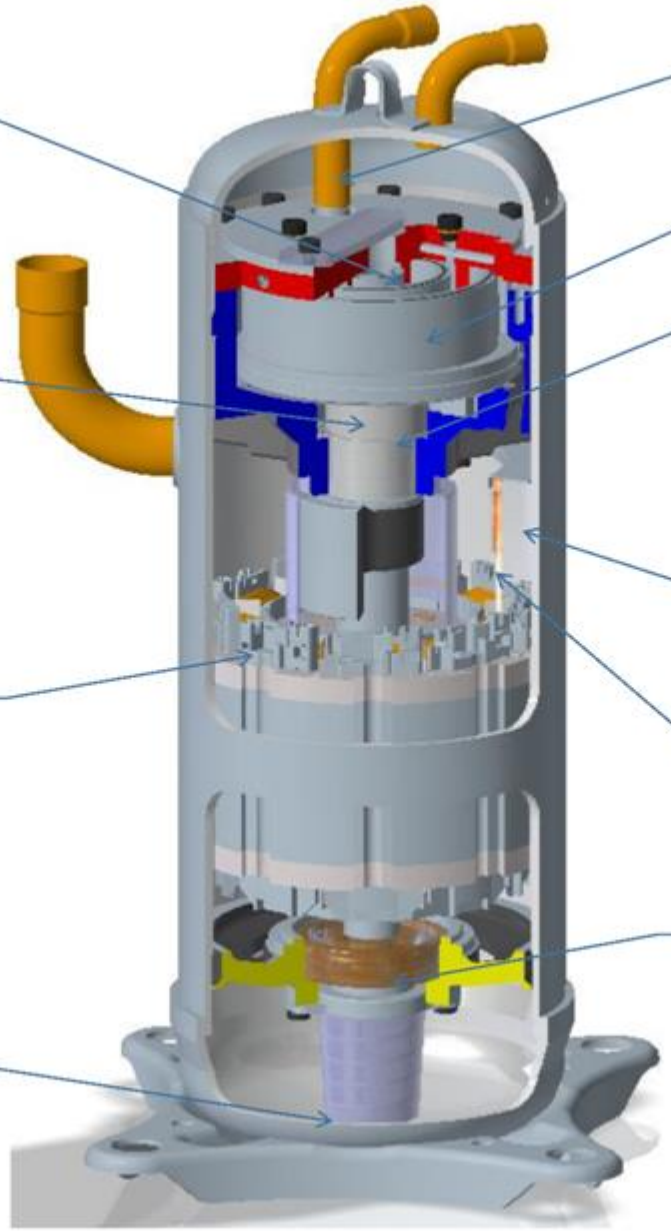
Highly efficient 6-stage split iron core permanent magnet DC motor

Efficient, high-quality, low-noise, miniaturized motors.



Oil tank self-cleaning design

Effectively absorb foreign matters and iron impurities in the circuit to improve the overall reliability.



Efficient EVI technology

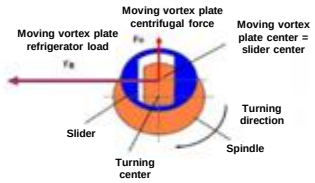
EVI tubes are large and durable enough to ensure increased EVI flow, higher capacity output and expanded operation.

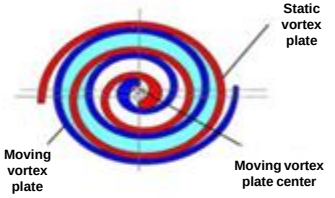
Aluminum moving vortex plate design

Reduce axle and axle sleeves load for high load and high speed correspondence, ensuring overall system reliability.

Unique radial swing design

Better engagement of moving and static vortex plates, less leakage, and dramatically improved reliability and efficiency.





Low-pressure cavity design

Effectively prevent liquid hammer and control the temperature rise of the motor and the core to realize larger operating range and to improve the life of the unit.



Oil return pipe in cavity design

Better control of oil circulation to ensure that the core is running with sufficient oil to guarantee the overall reliability

Gear oil pump with filter (volumetric)

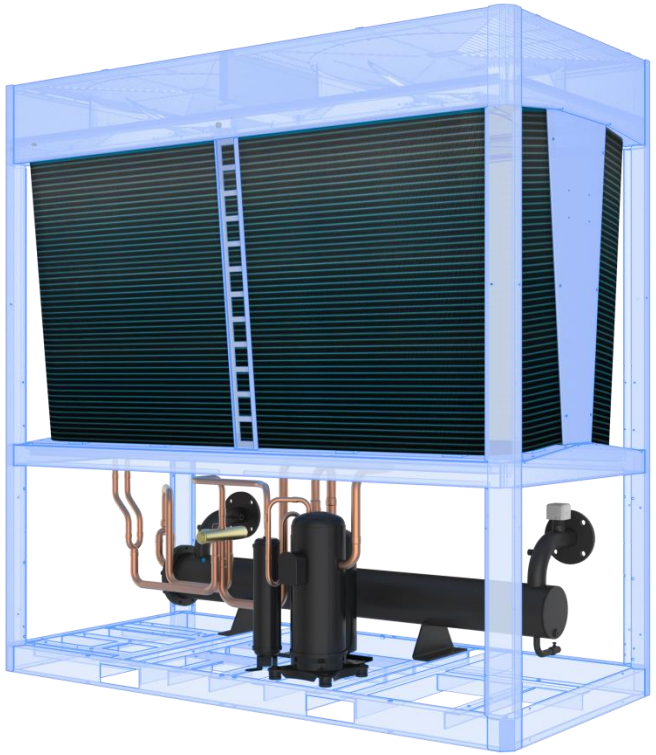
Use proactive fuel supply to ensure sufficient and clean oil at all times to improve the overall life.



Note: Smaller and 30% lighter than 12hp fixed speed unit

Advanced 25HP DC inverter scroll compressor

Single 65-module system design that is concise and reliable. (Patent 202020931775.1)



Disadvantages of **multiple** systems

Using 2 compressors in parallel results in uneven oil return and refrigerant distribution as well as **poor reliability**;

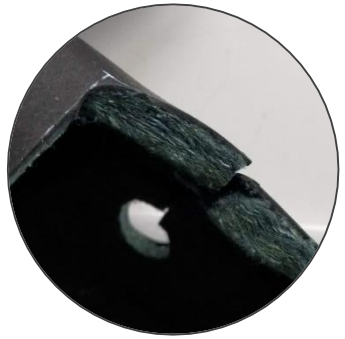
Using 2 standalone systems adds complication to the design and requires more parts, causing **high overlay failure rate**;

Advantages of **single** system: the number of copper pipes, valves, gas-liquid separators, liquid receivers, sensors, etc., is reduced by half, making the unit **simple yet reliable**

Patented quintuple sound insulation for compressor room

(Patent No. 202020931775.1)

1. Acoustic block



Green felt with butyl rubber material to block low to high frequency noises

2. Ultra-silence acoustic panel absorber



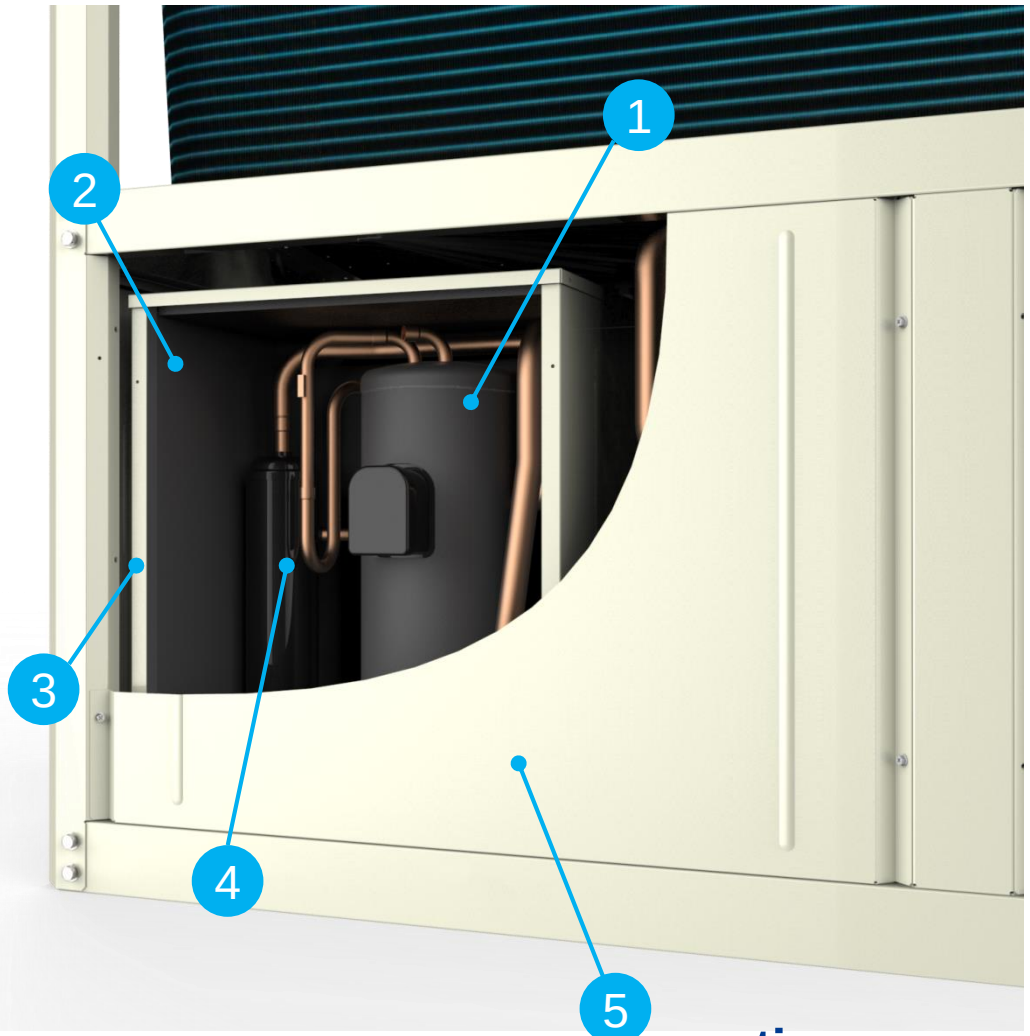
High-density foaming damping material to effectively absorb vibration and noise across frequencies



3. Hermetic soundproof box

4. Air suction and discharge copper pipe, oil separator, silencer, etc., placed in the compressor soundproof box

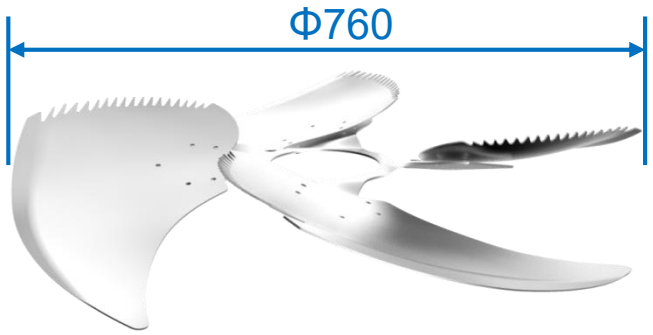
5. Protection panel around the unit to further block noise

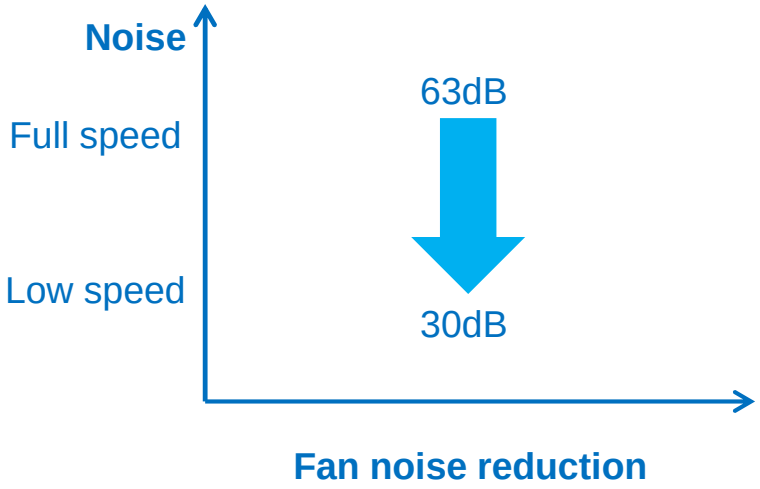


Industry's largest diameter axial fan

under equal air flow

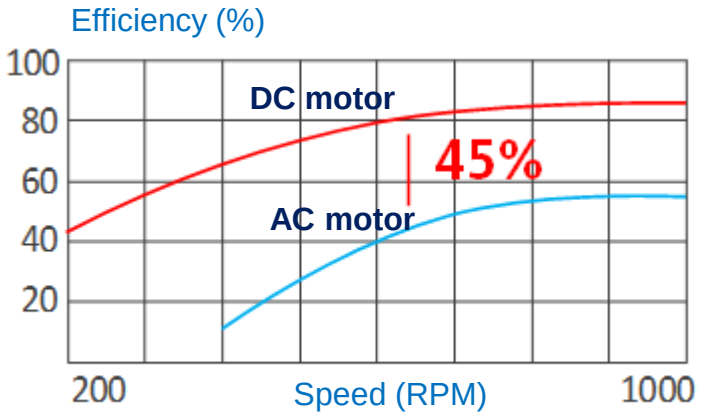
Sawtooth large impeller, low rotating speed, low fan speed, operating noise in low speed for dual blower as low as 30 dB.





The brushless full DC fan motor features stepless speed adjustment of 15-100% and more stable operation, achieving higher energy efficiency while reducing noises.



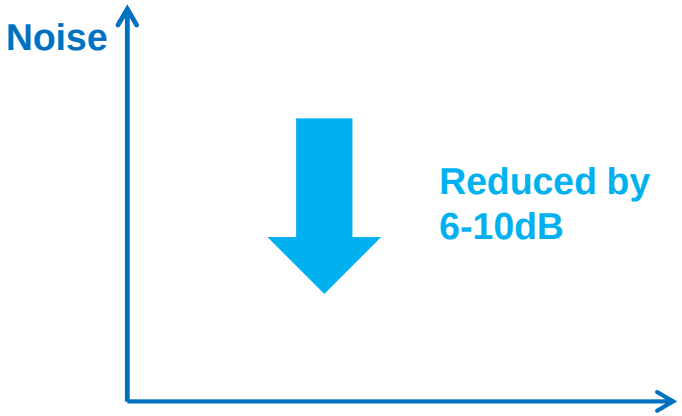


Frequency conversion and noise reduction

Three silent modes selectable

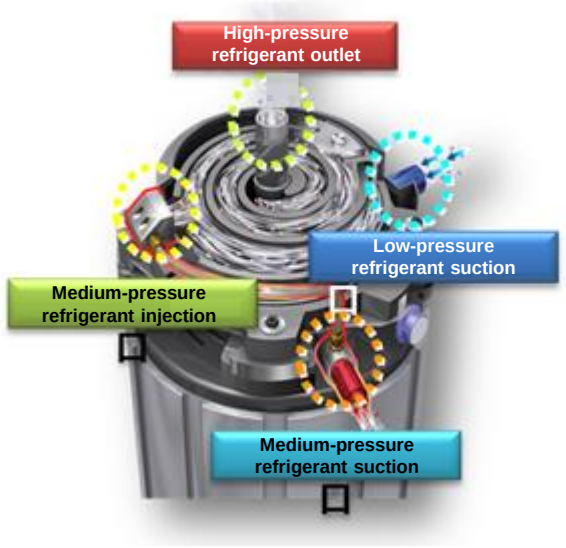
With wired controller settings, the unit automatically limits the operation frequency of the compressor and the fan and actively reduces noise to meet customers' higher requirements.



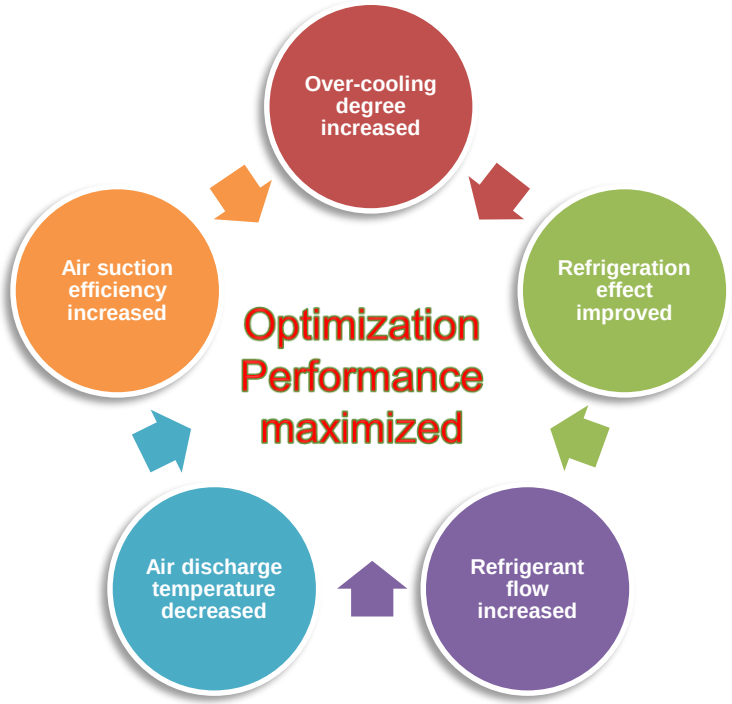


EVI design

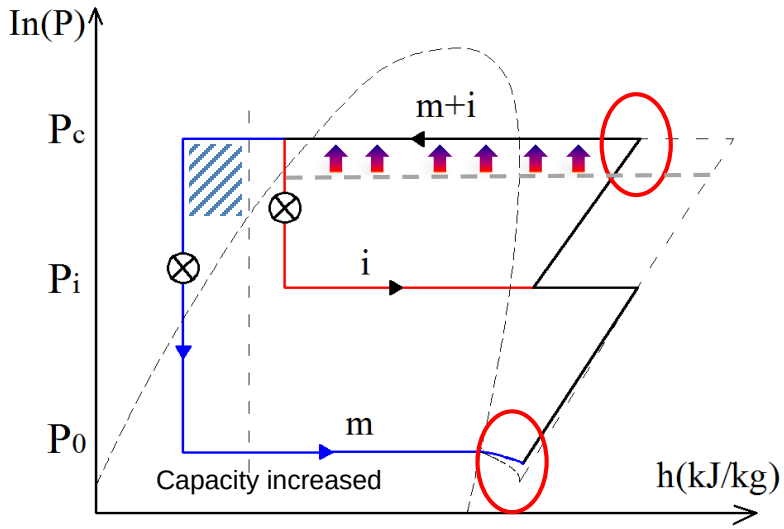
1. Operation process of EVI:



2. Actual effect of EVI:



3. Principle of EVI pressure-enthalpy diagram:

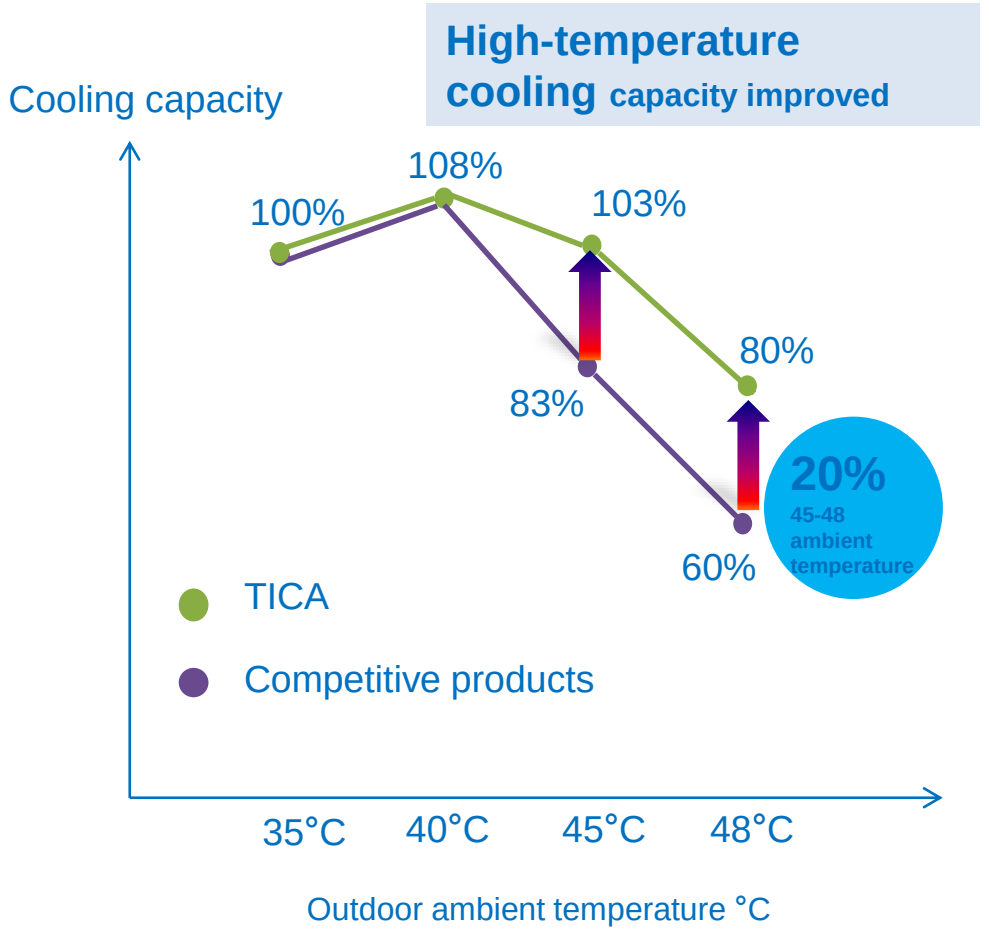


EVI design

Innovative use in **cooling**

When cooling at an temperature range of **45-55°C**, cool, gaseous refrigerant is injected to **reduce** air discharge temperature and improve compressor reliability

Designed with EVI economizer over-cooling circuit to **increase** primary circuit liquid refrigerant **over-cooling degree**, while discharge temperature is reduced to further allow **frequency rise** to increase primary circuit refrigerant flow with high-temperature cooling capacity **increased by 20%**



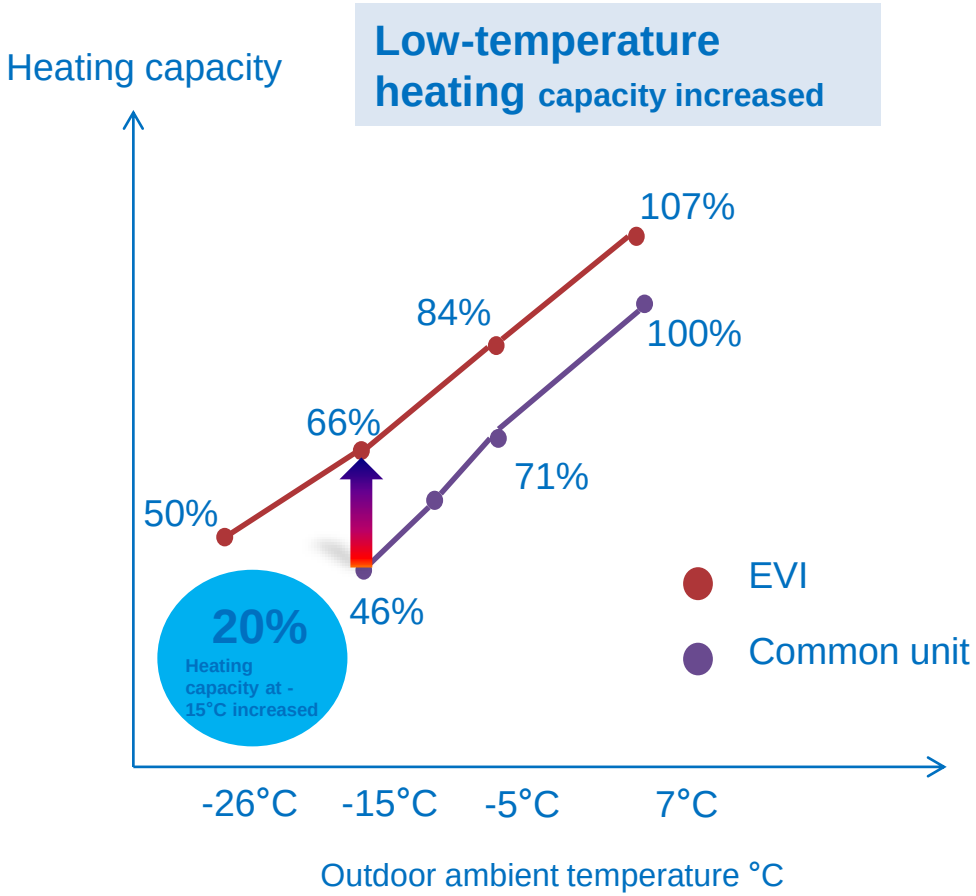
EVI design

Optimization **Heating**

Designed with EVI to **increase** jetting refrigerant **flow**, boost high pressure, increase heating capacity and water temperature

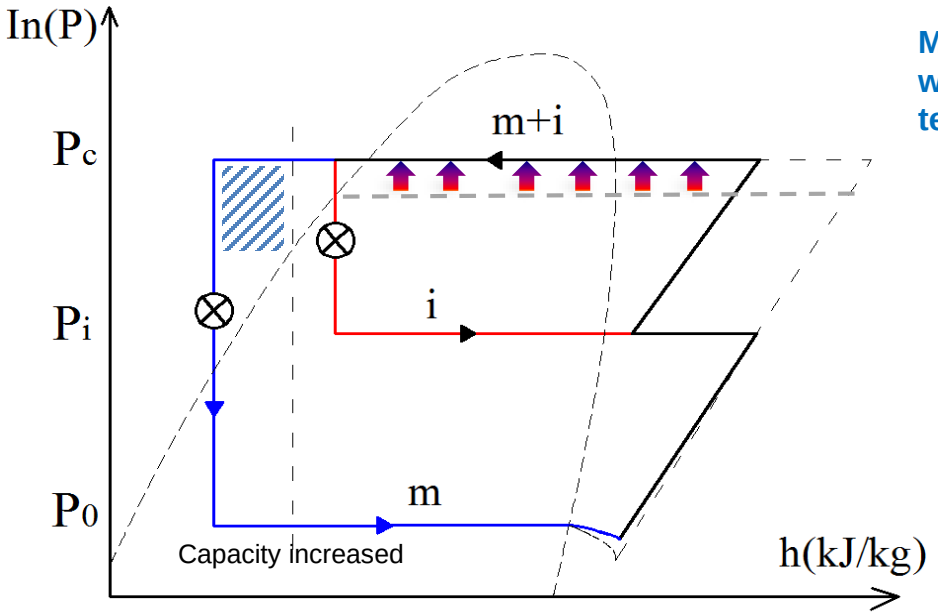
When heating at a temperature as low as **-15°C**, the middle air replenishment reduces the compression ratio and air discharge temperature, and increases refrigerant circulation with low-temperature heating capacity **increased by 20%**

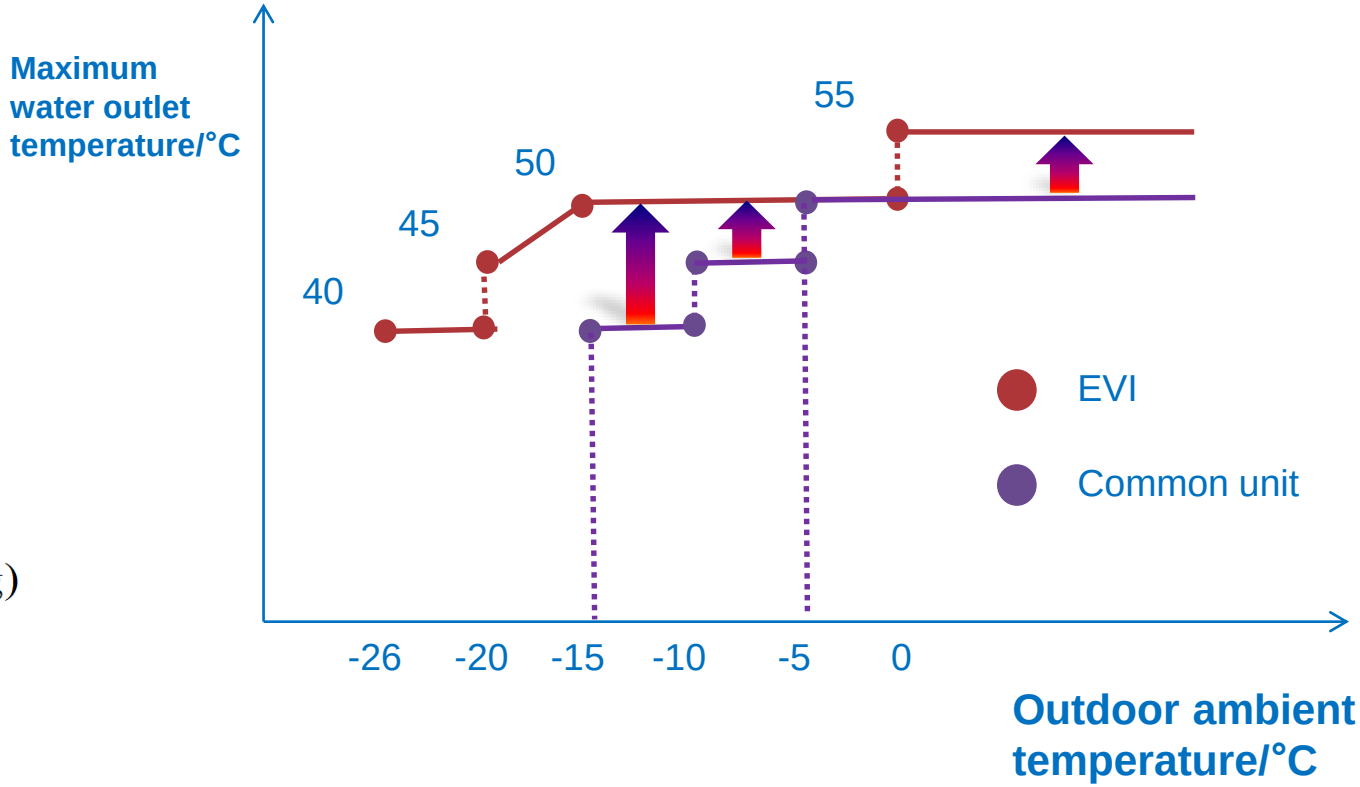
Heat even at an ambient temperature of **-26°C**



EVI design

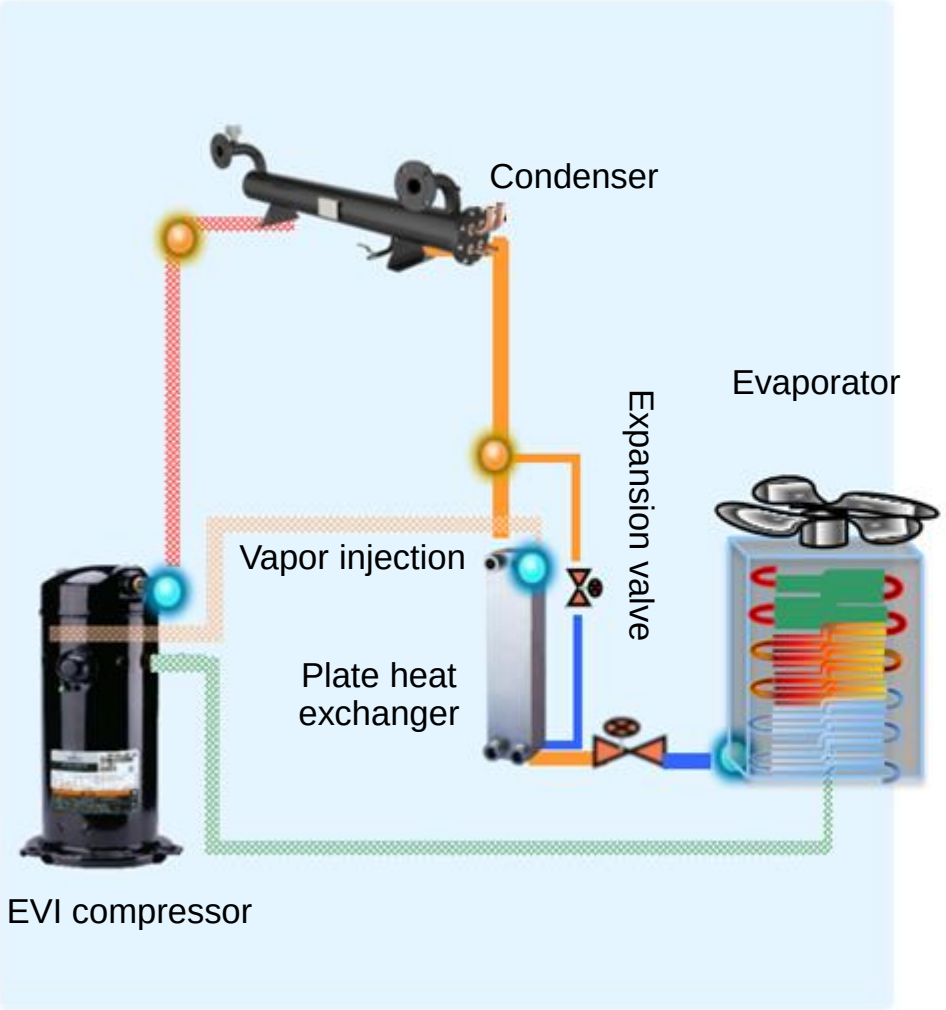
Designed with EVI to allow the compressor to operate at high pressure ratio
5-10°C increase in low-temperature heating water temperature to meet the needs of floor heating, fan coil unit, and new style radiator



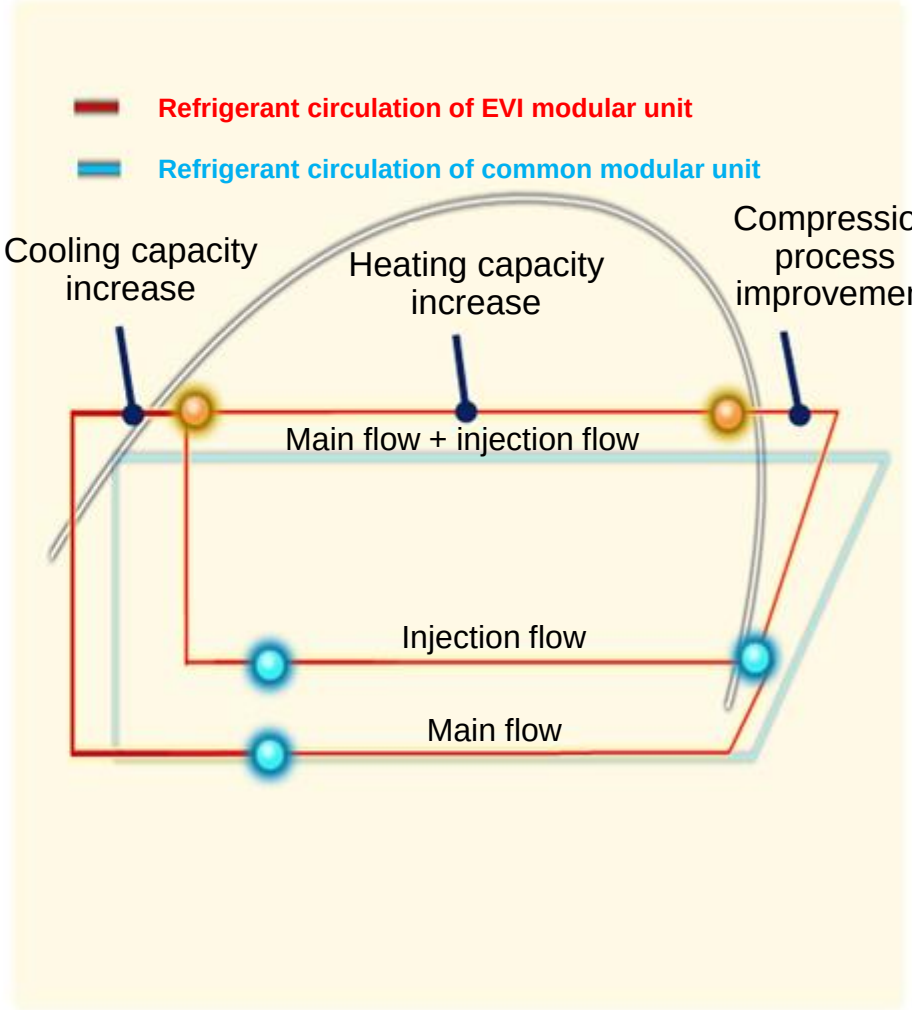


EVI design - system schematic diagram

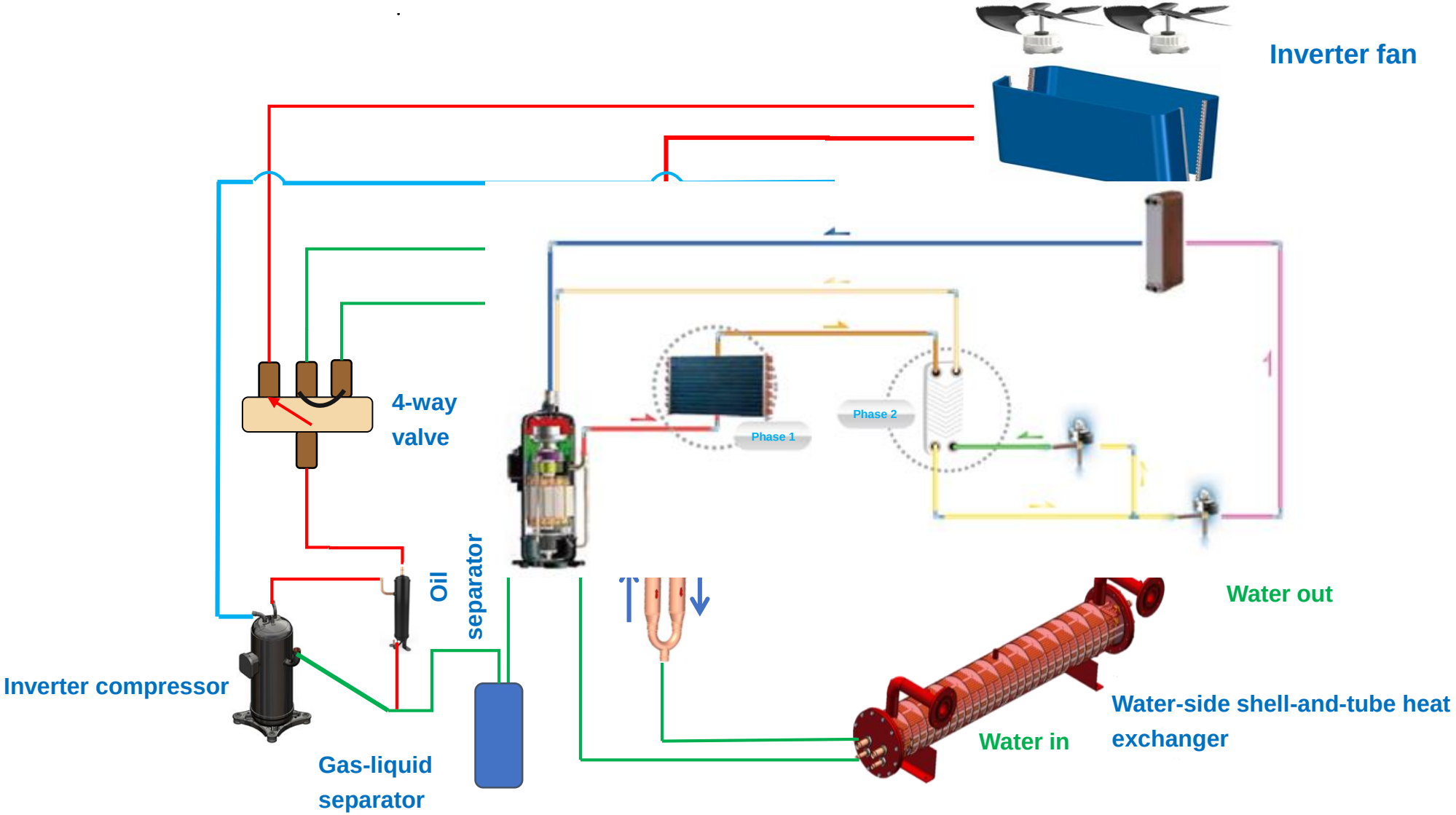
EVI circulation schematic diagram



EVI circulation pressure-enthalpy diagram

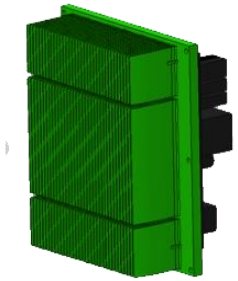
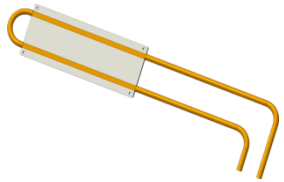
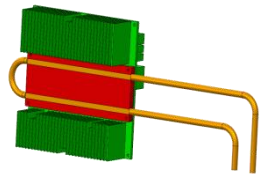
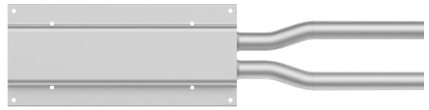


EVI design - system principles



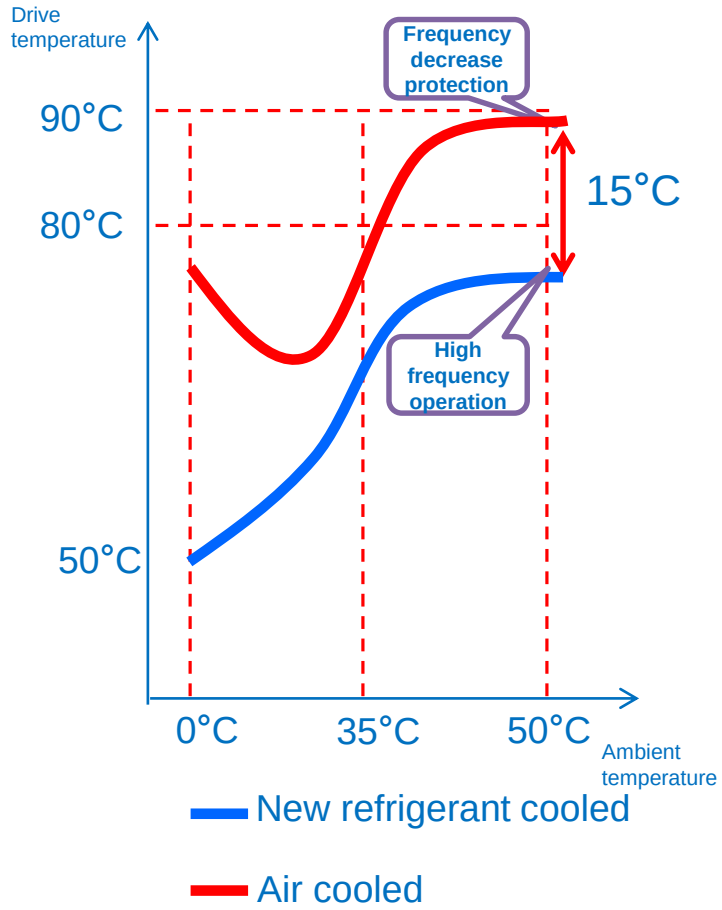
What is the main difference from fix-speed TCA201XHE?

New drive cooling Compressor driven

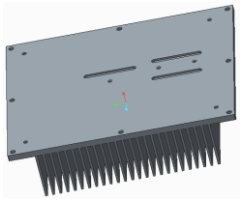
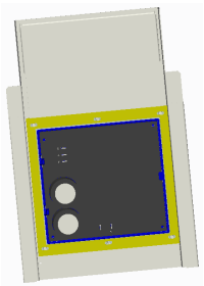
Heat dissipation mode	Air-cooled	Refrigerant cooled	Refrigerant cooled + air cooled	New refrigerant cooled
Figure				
Notes:	Poor heat dissipation	Good heat dissipation Affected by refrigerant volume	Good and stable heat dissipation	Refrigerant cools directly inside the heat sink for better heat dissipation

New cooling schemes to significantly improve compressor operation frequency and improve unit capability

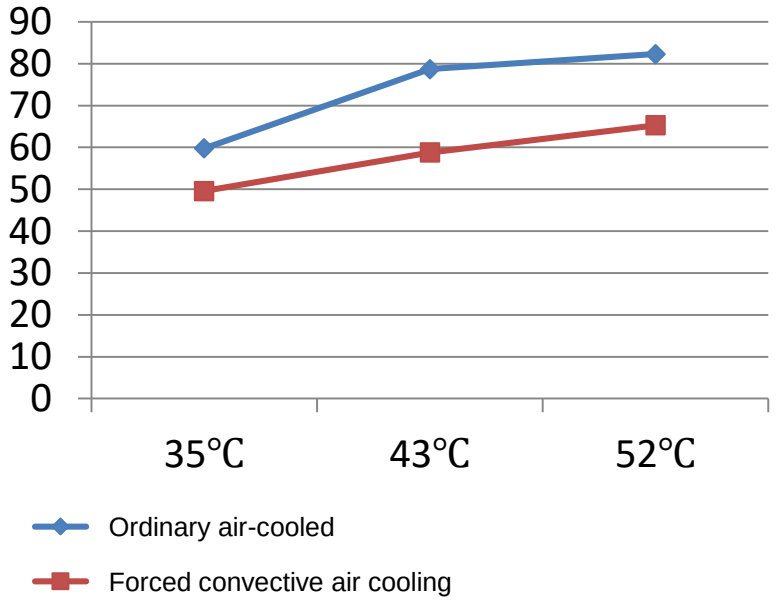
Patent: CN201621167898



New drive cooling Fan driven

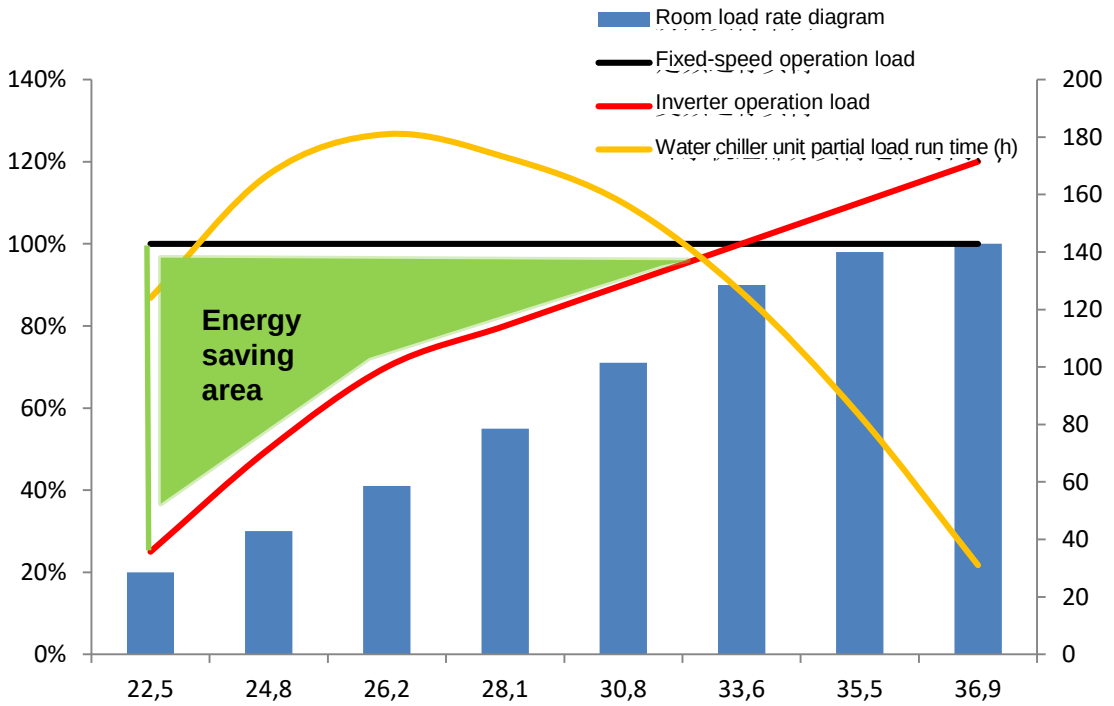
Heat dissipation mode	Ordinary air-cooled	Forced convective air cooling
Figure		
Notes:	Poor heat dissipation	Designed with air convection channels Drive **temperature drop up to 20K**

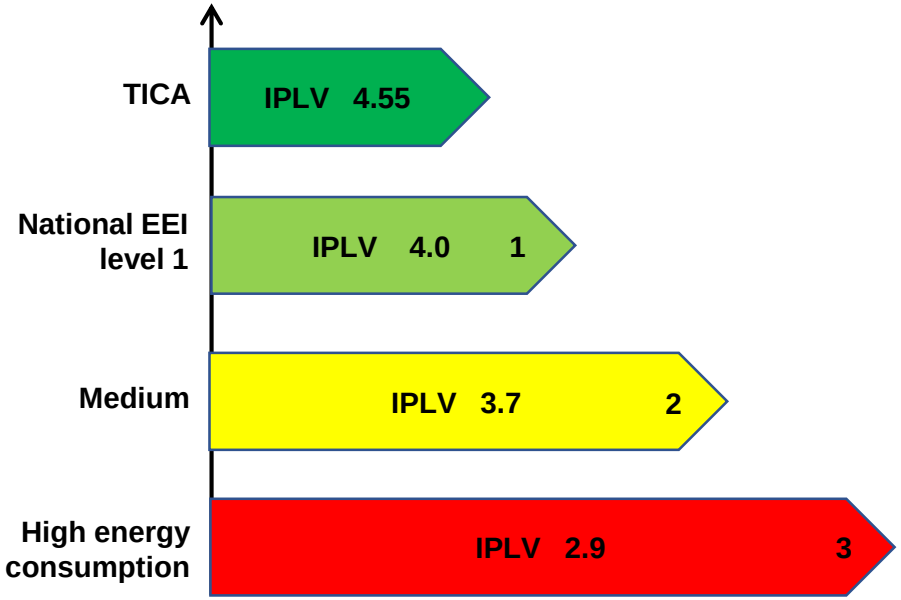
Patent: 202010137861.X



IPLV dual grade-1 energy efficiency

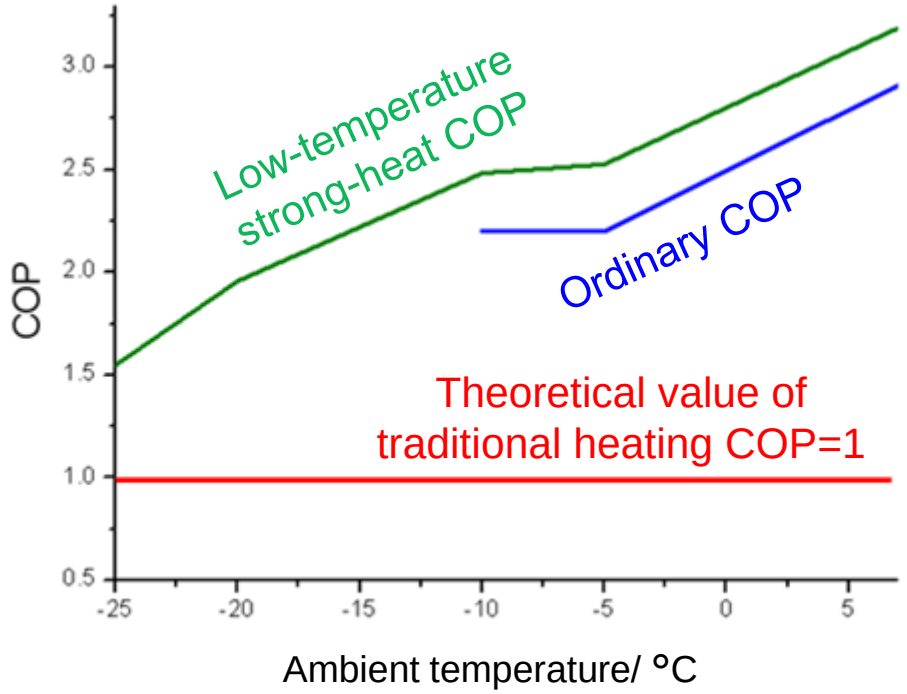
Adopt full DC inverter, secondary over-cooling of EVI economizer, and combine with independently developed efficient and reliable inverter drive and optimal energy efficiency control program to achieve **cooling IPLV up to 4.55**, far beyond the national EEI level 1 of GB 19577 standard. Compared with fixed speed units, IPLV is greatly improved by **25%**.

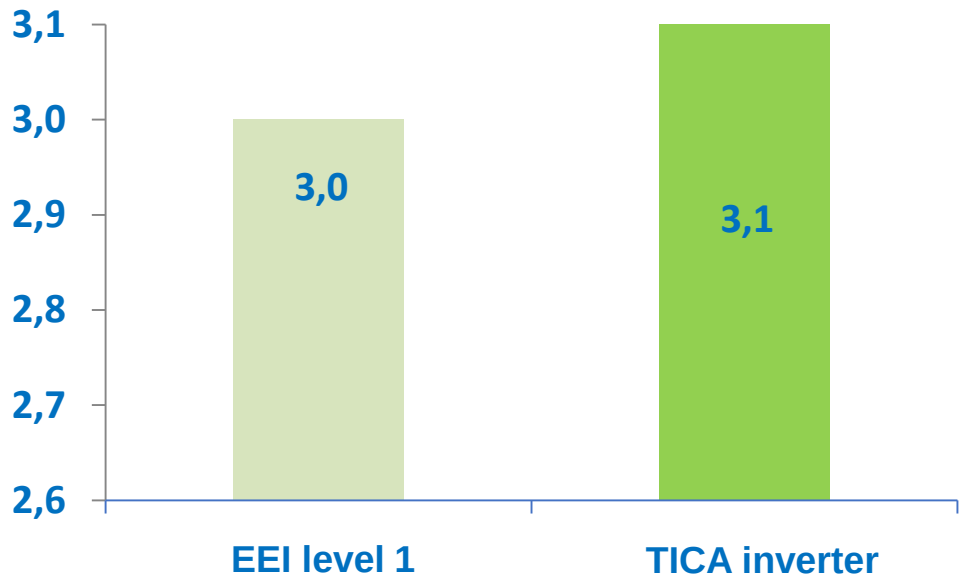




IPLV dual grade-1 energy efficiency

The unit can reliably heat the room at -26°C. **Heating IPLV is as high as 3.1**, meeting the national EEI level 1 for heating of new GB 37480-2019 standard.



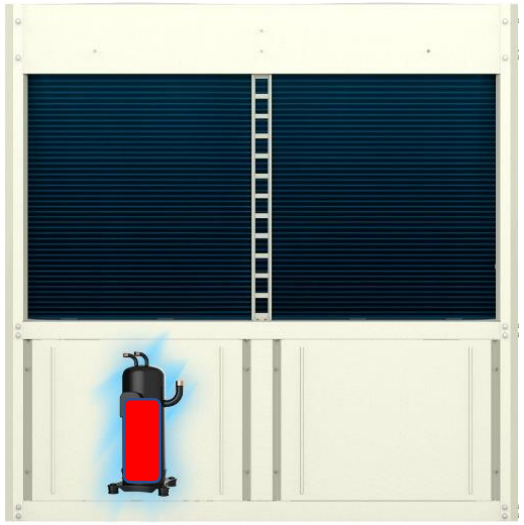


IPLV dual grade-1 energy efficiency

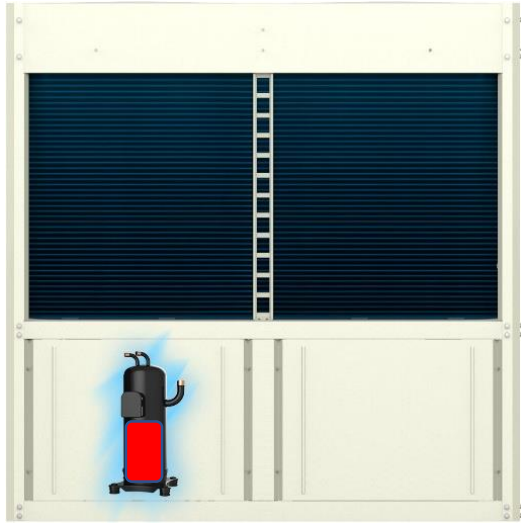
Partial load operation prioritized:

When multiple modules are combined, the frequency of each compressor is intelligently controlled, with priority given to the operation in an energy-efficient partial load area.

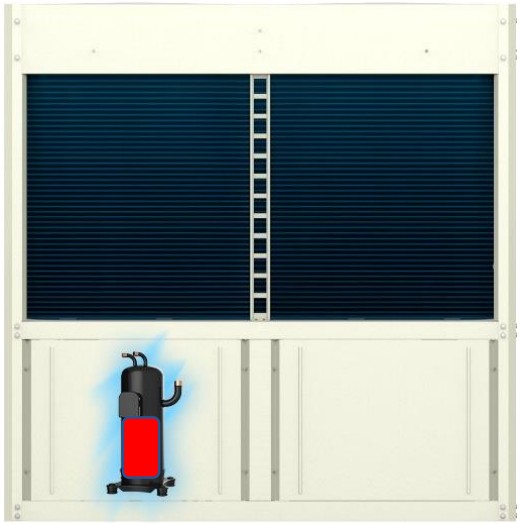
At the same time, compressor running time and load are balanced to facilitate uniform wear of each module compressor.



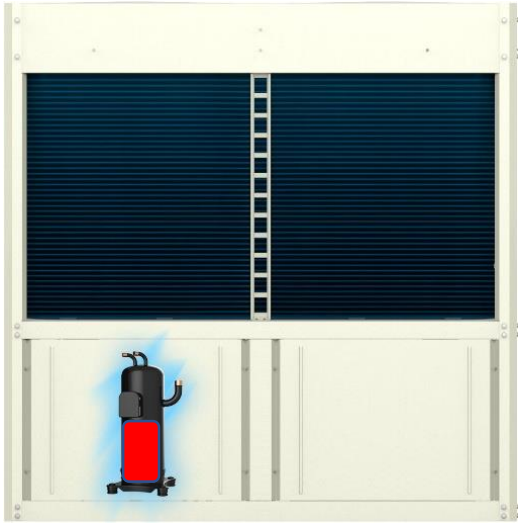
60% load



50% load

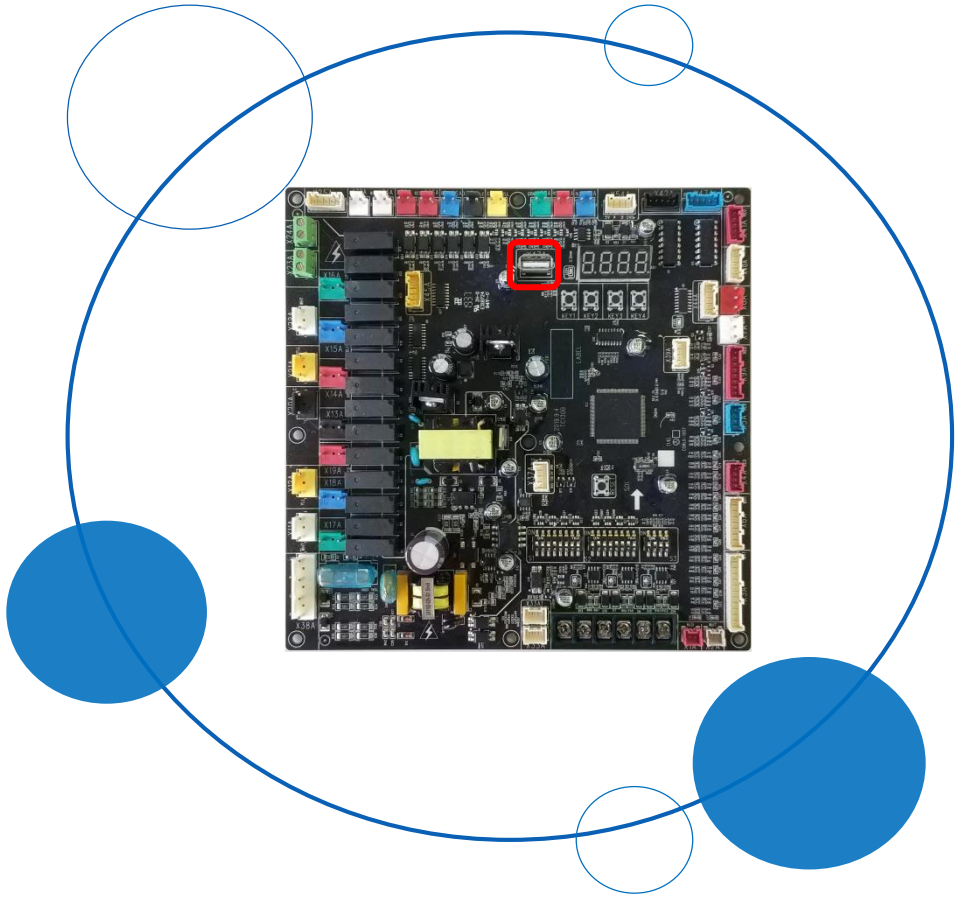


50% load



50% load

Intelligent control - black box





USB

Easy after-sales **fault analysis**

Convenient **program upgrade**

Ultra-large storage space

Data kept for **ten years**

Two touch controllers available

Meet different customer



Standard



7-inch color touch controller (optional)

New functions added: silent mode, auxiliary electric heating forced enabling, and submodule disabling.

Intelligent combination mechanism
fixed speed + inverter



Free combination of up to 16 variable and fixed speed modules

Meet project requirements such as energy-saving transformation









V-FORCE features



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V-FORCE

Product features:

Operate under all conditions

Operate at -26°C-55°C
Performance improved by 20% at extreme conditions

Full inverter energy saving

Dual grade-1 IPLV for cooling and heating
IPLV comprehensive energy saving rate up to 26%

Various application scenarios

Solutions to meet normal household use, special process and sanitary conditions, cooling only, low temperature and strong heat, etc.
Full application scenario alternative

Streamlined appearance design

Full concealed design

Four-way air return

Simple system architecture

Single compressor design

Refrigerant line optimization to reduce solder joints

Easy user experience

Full series compatibility of modular unit
User-friendly control panel with one key operation
Environmental friendliness with multiple silence

PART 2

Comparison with Competitive Products

Comparison with Competitive Products - Overview

Name	MQ		YORK	MD	TICA	
	MAC-E	Hexagonal air-cooled inverter heat pump unit	X-Power series YVAE-X efficient premium version	Air-cooled heat pump inverter module unit I series	V-FORCE full-operating condition inverter	
Appearance						
Model	MAC230ER5/MAC450ER5	MIC340AR5(100kW)	YVAE065/100/130XRME	LSQRFL30/60/90M/VN1-I	TCAV035/65BHE	
Cooling capacity (kW)	65/130	100kW	58.5/98/125	30/58/90	33.5/65	
Heating capacity (kW)	70/140	?	60/99/127	33/64/100	34/75	
Cooling power input	21.5/41.9	?	19.8/32.2/39.9	9.84/19.33/29.8	12/21.2	
Heating power input	22/43.7	?	19.7/31.8/45	10.3/20/31.45	10.5/23.4	
Cooling COP	3.02/3.1	3.20	2.95/3.04/3.13	3.05/3/3.02	2.79/3.07	
Heating COP	3.18/3.2	3.35	3.05/3.11/2.82	3.2/3.2/3.18	3.24/3.2	√
Performance specifications of partial load	4.51/4.42	4.80	4.65/4.39/4.48	4.6/4.49/4.53	4.6/4.55	√
Refrigerant	R410A	?	R410A	R410A	R410A	
Cooling ambient temperature range (°C)	-15 - 52	?	-17.8 - 48	-10-50	-20 - 55	√
Heating ambient temperature range (°C)	-26 - 30	-30-?	-15 - 43	-15 - 30 -20-30 (90)	-26 - 55	√

Comparison with Competitive Products - Overview

Name		MQ		YORK	MD	TICA	
		MAC-E	Hexagonal air-cooled inverter heat pump unit	X-Power series YVAE-X efficient premium version	Air-cooled heat pump inverter module unit I series	V-FORCE full-operating condition inverter	
Appearance							
Model		MAC230ER5/MAC450ER5	MIC340AR5(100kW)	YVAE065/100/130XRME	LSQRFL30/60/90M/VN1-I	TCAV035/65BHE	
Compressor	Type	DC inverter scroll type + EVI (Copeland)	?	Efficient scroll type DC inverter (Inverter from Hitachi + fixed speed from Copeland)	30/60 inverter dual rotor compressor 90 full DC inverter EVI (Copeland)	Scroll type DC inverter EVI (Inverter from Mitsubishi)	√
	Capacity control	15% to 100%	?	Single module 12.5%-100% stepless capacity control	25% to 100%	15% to 100%	
	Qty	2	?	2/3/4	1/1/2?	1	
Water-side heat exchanger		Braze welding plate heat exchanger	?	Braze welding plate heat exchanger	Efficient plate heat exchanger	High-efficiency shell-and-tube heat exchanger	
Fan	Qty	2	?	1/2/2	1/2/3	1	√
	Type	Inverter axial flow low noise	?	Brushless DC inverter	Full-DC	DC low-noise axial flow fan	
Combined modules		16	8	16 (standard centralized controller) 32 (Smart View centralized controller)	16	16	
Noise dB(A)		49-67 51-69	?	68.4/69/70	?		
		The speed is lowered automatically at low load operation. The operating noise of the unit can be as low as 49 dB(A)/51 dB(A)	?	In silent mode, the operating noise can be reduced by 5-7 dB(A) when the fan is operating at low speed	The noise can be lowered by 4 dB(A) in silent mode, and 8 dB(A) in extra silent mode	The noise can be as low as 50 dB(A) in partial load	
Combined with fixed speed modules		Yes	?	Yes	No	Yes	
Options		?	?	Baffle & grille/noise reduction assembly Smart View centralized controller/standard centralized controller/SC-Equip assembly (Convert BACnet)/inverter hydraulic module	Optional hydraulic module	/	

Comparison with competitive products - MQ Compressor

Variable volume ratio vortex plate

- Variable volume ratio vortex plate is specifically optimized for cooling and heating performance, effectively improving overall system efficiency.

Radial compliance technology

- Reduce radial leakage at low vortex speeds and improve low-speed energy efficiency of the overall system.

Efficient internal oil ring design

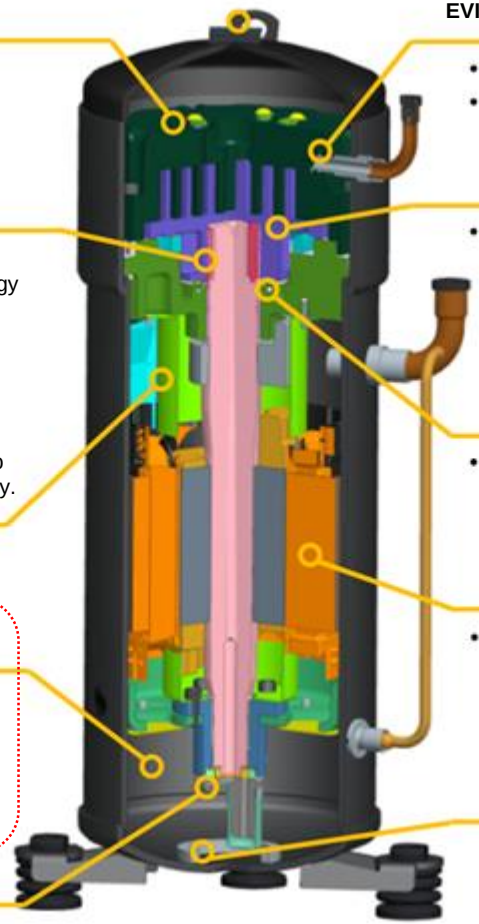
- Reduce the oil spill rate at the vent to increase the heat exchange efficiency.

Full high-pressure cavity design

- Eliminate harmful suction superheat degree to improve vortex compression efficiency;
- Provide a large silencing cavity to reduce the overall compressor noise level.

Volumetric oil pump

- Provide stable and reliable fuel supply to the bearing system;



EVI technology

- Better control of air discharge
- temperature;
  Raise low-temperature heating capacity.

Unique oil sealing technology

- Reduce leakage during vortex compression to boost overall energy efficiency;

Axial compliance technology

- Reduce axial vortex leakage to improve overall energy efficiency.

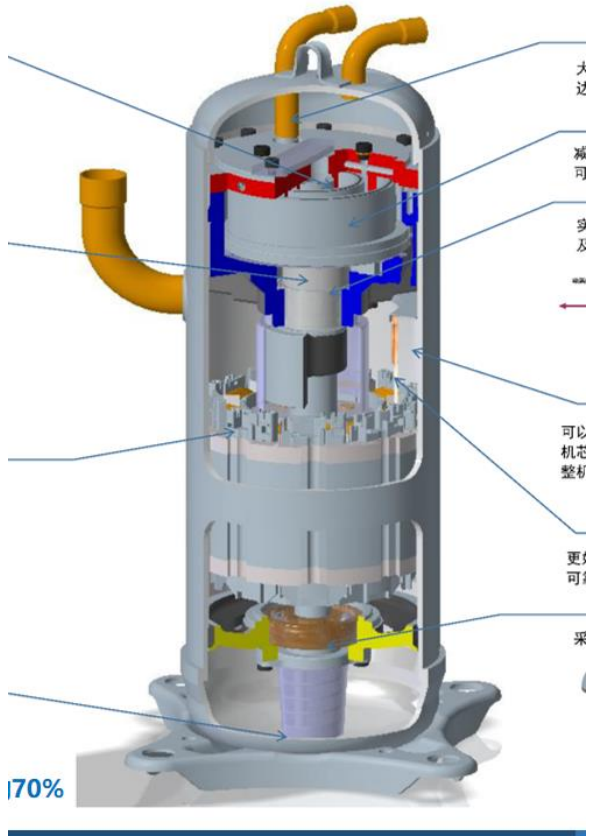
Efficient permanent magnet DC motor

- Provide higher motor efficiency to improve overall energy efficiency.

Oil tank self-cleaning design

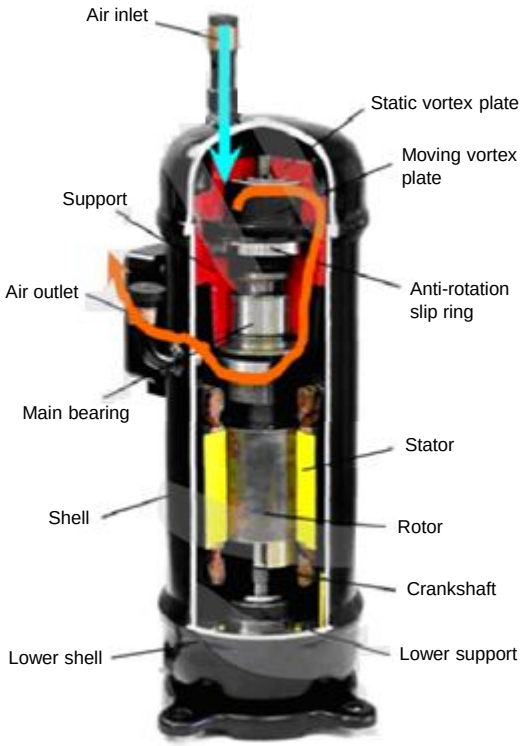
- Avoid damage to moving parts caused by small impurities.

High pressure cavity compressor

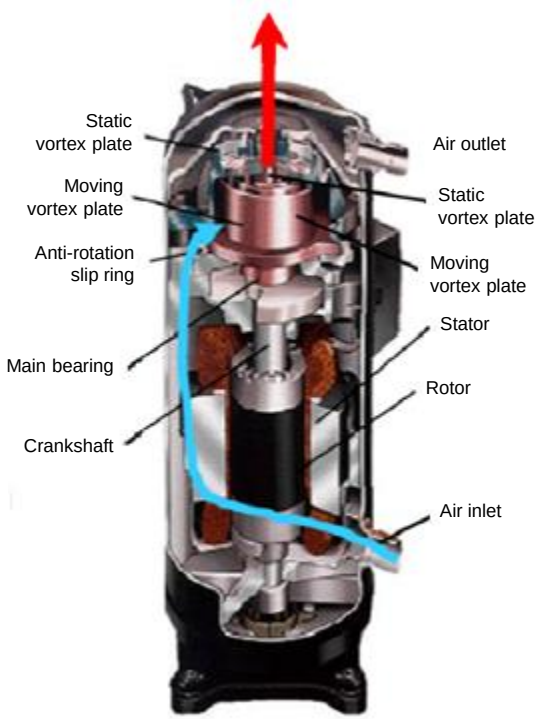


Low pressure cavity compressor

Comparison with competitive products - MQ Compressor



High pressure cavity compressor

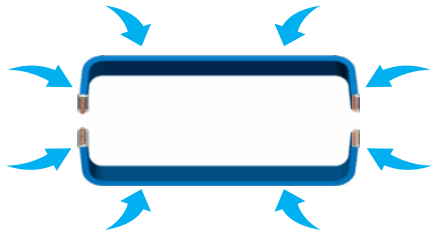
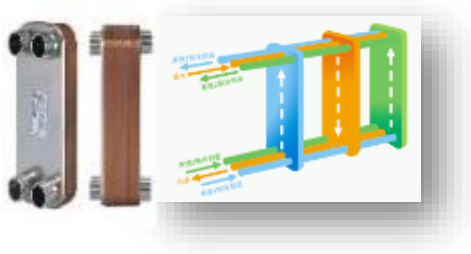
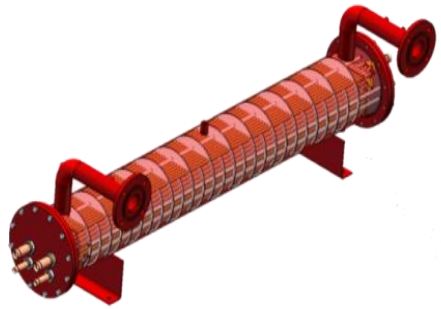


Low pressure cavity compressor

Compressor form	Advantages	Disadvantages
High pressure cavity	1. Direct air suction basically does not dilute lubricating oil; 2. Good oil balance effect; 3. Large air discharge buffer volume, small vibration, and uniform gas delivery	1. Motor operating environment temperature is high; 2. Oil is pumped using differential pressure; 3. The compressor is easily damaged by impurities and foreign matters; 4. Small air suction buffer volume and poor air suction silence effect
Low pressure cavity	1. The motor is cooled directly with low temperature return air, and the cooling effect is good. 2. Most parts of the shell bear low pressure, and air tightness and load-carrying capability are good; 3. Strong resistance to liquid hammer as well as foreign matters and impurities; 4. Oil is pumped using an oil pump	1. It is highly possible that the return liquid may dilute lubricating oil; 2. It is hard to achieve oil balance; 3. Strong air suction preheating reduces volume efficiency; 4. Small air discharge buffer volume with large noise and vibration

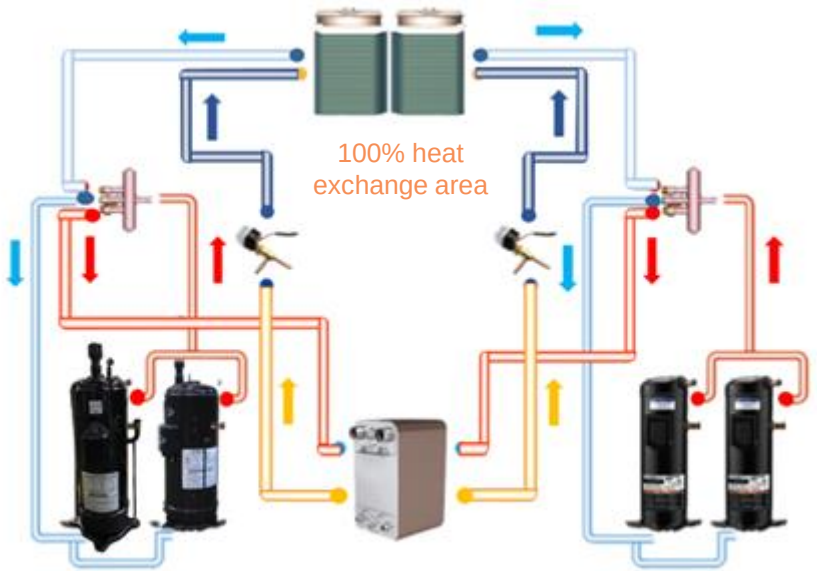
✡Whether low pressure cavity compressor can be jointly submitted for bidding?

Comparison with competitive products - MQ key component difference

Key components	MQ	TICA	Content of discussion
Air-side heat exchanger	Semi-M-type heat exchanger 65 Ring-shaped heat exchanger 130 	Dual U heat exchanger 	1. The air resistance is uniform at all sides and the heat exchange effect is good. 2. Inclination angle design results in better drainage effect in defrosting process.
Water-side heat exchanger	Braze welding plate heat exchanger 	Shell-and-tube heat exchanger 	1. Strong antifreezing capability; 2. Strong anti-fouling capability.

Comparison with competitive products - YK Compressor



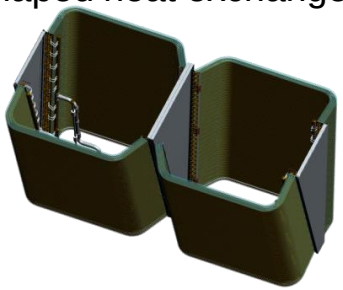
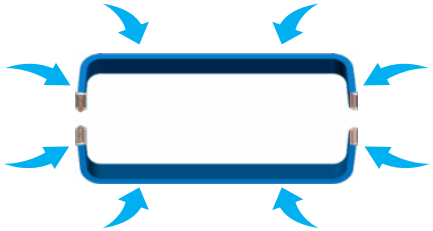
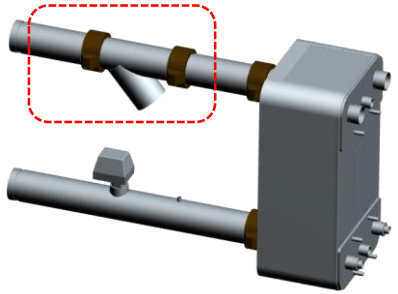
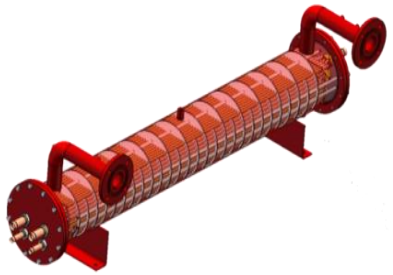
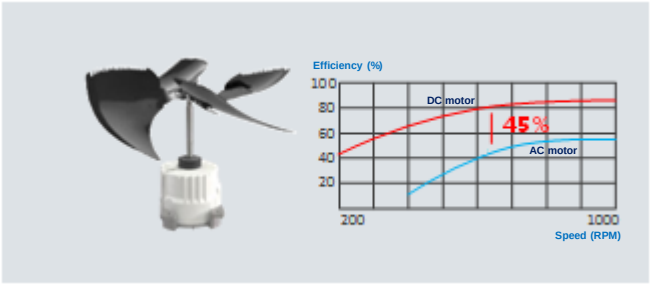


For example, York 130 unit uses:
1 Hitachi inverter + 1 Hitachi fixed speed unit + 2 Copeland fixed speed units

YORK
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- There are two different types of compressors in a unit, resulting in poor interchangeability of accessory stock;
- The fixed speed compressors are used as major compressors, so the inverter response speed is slow, and the control is needed to respond to load change;
- There are too many compressors, and every two compressors are connected in parallel to one system, leading to poor oil balance and stability.

Comparison with competitive products - YK key component difference

Key components	YK	TICA	Content of discussion
Air-side heat exchanger	U-shaped heat exchanger 	Dual U heat exchanger 	Basically the same
Water-side heat exchanger	Braze welding plate heat exchanger 	Shell-and-tube heat exchanger 	1. Strong antifreezing capability; 2. Strong anti-fouling capability. (York is equipped with standard 50-mesh built-in water filter)
Fan	EC fan 	Inverter fan 	Basically the same

Comparison with competitive products - YK key precautions

1. Non-stop defrosting



Upgrade of defrosting

- Simultaneous defrosting and heating via single module: With innovative defrosting design, single-module dual-refrigerant system operates in a fully independent way, enabling defrosting and heating at the same time via a single module and ensuring reliable heating in winter; fixed speed design translates into quicker defrosting.
- Non-stop heating & defrosting: When multiple modules are combined, single module defrosting does not affect the heating operation of other modules, and each module can realize defrosting and heating at the same time; therefore, the heating effect of the whole system can be guaranteed.

2. Mutual backup



Mutual backup of refrigerant systems

- Single-module dual-refrigerant system operates in a fully independent way; therefore, any failure on one refrigerant system does not affect the normal operation of the other system, making the standalone system more reliable;

☆York system is a dual system, which means it can realize the above two features. It is difficult for TICA products to achieve the two.

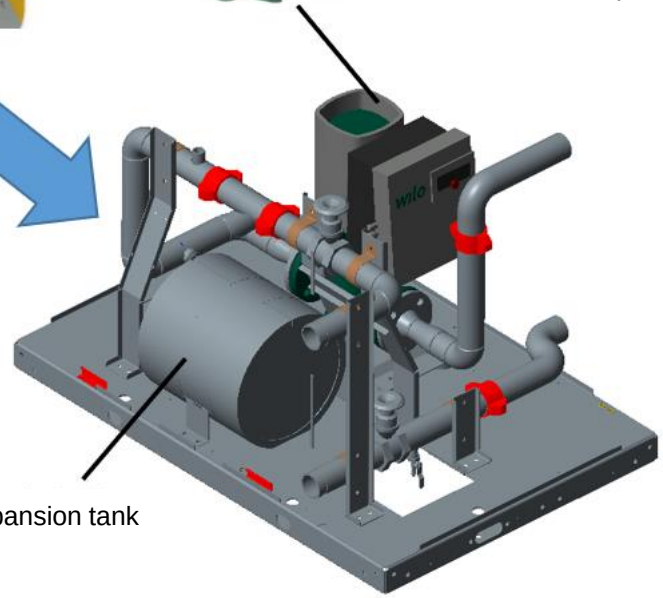
Comparison with competitive products - YK Key precautions



Hydraulic module box



GIGA inverter water pump



Expansion tank

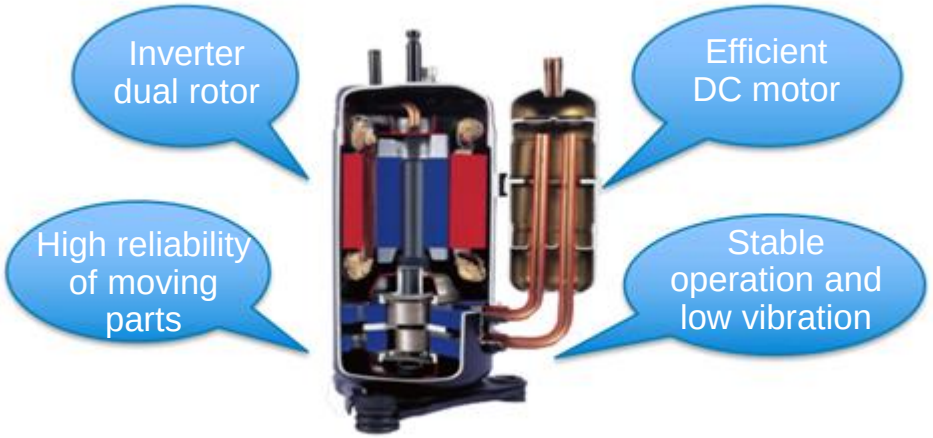
YORK

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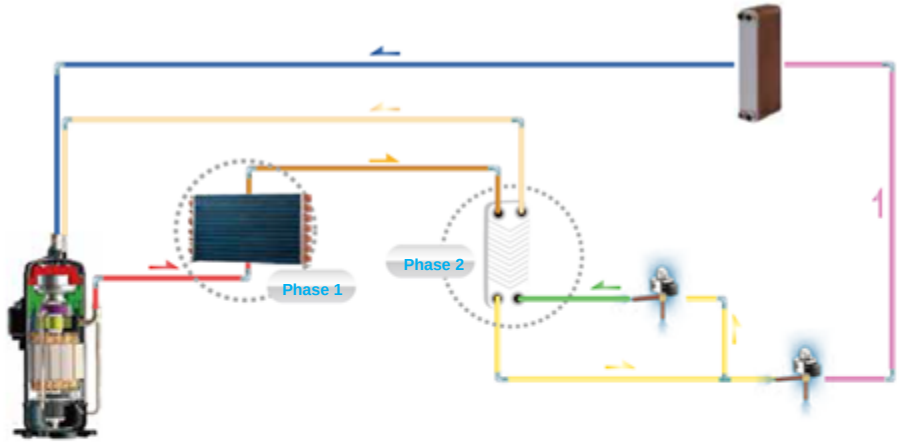
York's unit is also available with options like external inverter hydraulic module

- It does not apply to all situations due to large differences in the lift;
- Factory installation can be selected, which means inconvenient transportation and lifting; there is no value if installed on site;
- TICA is also developing external hydraulic modules. Please consult us if you have any demand.

Comparison with competitive products - MD Compressor



LSQRFL30M/VN1-I and LSQRFL60M/VN1-I use inverter dual-rotor compressor that provides 25%-100% stepless regulation to allow capacity to be changed according to load. They do not need to be switched on and off frequently, ensuring comfort and energy efficiency.



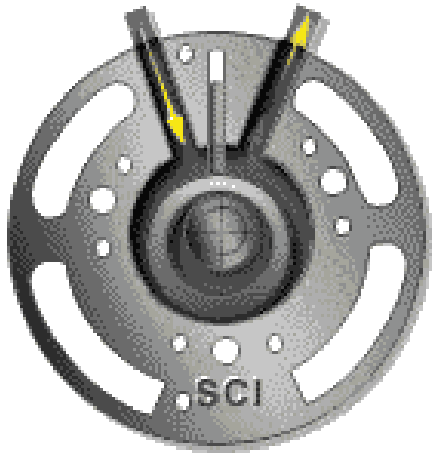
All DC inverter EVI compressor used in LSQWRFL90M/VN1-I

- Keep running stably at -20 °C
- Outlet water still reaching 45 °C at -15 °C

- ✯TICA 65 module VS Midea 60 module: cooling capacity/IPLV/extreme condition operation and other indexes outperform Midea;
- ✯The rotor compressor is slightly lower than the scroll compressor in performance;
- ✯The 90 compressor is the same as the MQ.

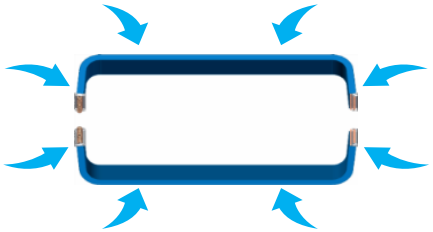
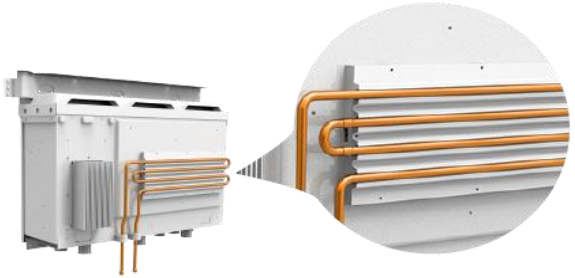
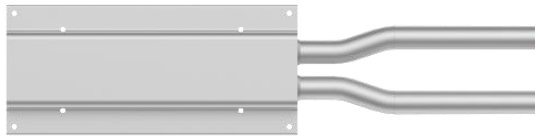
Comparison with competitive products - MD Compressor





Scroll compressor	Rotor compressor
- third generation compressor	- second generation compressor
- continuous air suction compression, achieving high efficiency	- single air suction and compression
- strong heating capacity	- relatively poorer heating capacity
- stable operation and long service life	- discharge valve slide is a quick-wear part
- high manufacturing costs	- low price and easy processing

Comparison with competitive products - MD Key component difference

Key components	Midea	TICA	Content of discussion
Air-side heat exchanger	I-shaped heat exchanger 	Dual U heat exchanger 	1. Small heat exchanging area; 2. If extra rows are added, air resistance will be increased; or if length is increased, more floor space will be needed.
Water-side heat exchanger	Braze welding plate heat exchanger	Shell-and-tube heat exchanger	1. Strong antifreezing capability; 2. Strong anti-fouling capability.
Dimensions	30: 1870*1000*1175/1.87 60: 2220*1055*1325/2.34 90: 3220*1095*1513/3.52	35: 1170×850×1695/0.99 65: 2000×950×2020/1.9	Compared with module 65, Midea product occupies 23% more area.
Heat dissipation of driver module	Heat dissipation via electrical control on refrigerant 	Cooling with refrigerant 	TICA refrigerant cools directly inside the heat sink for better heat dissipation

Comparison with Competitive Products - Review

Name	MQ	YORK	MD	TICA
	MAC-E	X-Power series YVAE-X efficient premium version	Air-cooled heat pump inverter module unit I series	V-FORCE full-operating condition inverter
Appearance				
Cooling ambient temperature range (°C)	-15 - 52	-17.8 - 48	-10-50	-20 - 55
Heating ambient temperature range (°C)	-26 - 30	-15 - 43	-15 - 30 -20-30 (90)	-26 - 55
Compressor	DC inverter scroll type + EVI (Copeland)	Efficient scroll type DC inverter (Inverter from Hitachi + fixed speed from Copeland)	30/60 inverter dual rotor compressor 90 full DC inverter EVI (Copeland)	Scroll type DC inverter EVI (Inverter from Mitsubishi)
Water-side heat exchanger	Braze welding plate heat exchanger	Braze welding plate heat exchanger	Efficient plate heat exchanger	High-efficiency shell-and-tube heat exchanger
Air-side heat exchanger	Semi-M-type heat exchanger 65 Ring-shaped heat exchanger 130	U-shaped heat exchanger	I-shaped heat exchanger	U-shaped heat exchanger
Cooling driven by	No description (air cooled)	No description (air cooled)	Heat dissipation with refrigerant	Heat dissipation with refrigerant
Floor area	1990*840*2440/1.67 2100*1100*2360/2.31	1500*1200*2440/1.8 2240*1200*2440/2.69 2240*1200*2440/2.69	30: 1870*1000*1175/1.87 60: 2220*1055*1325/2.34 90: 3220*1095*1513/3.52	35: 1170×850×1695/0.99 65: 2000×950×2020/1.9



TICA

We strive for excellence!

TICA SNG LLC

Tel.: +7 495 127 79 00,
+7 915 650 85 85,
+7 915 190 85 85

E-mail: info@tica.pro

www.tica.pro